

Course information

Vladislav Goncharenko
Radoslav Neychev



ITMO, fall 2026

Outline

1. Team
2. How to learn?
3. Prerequisites
4. Communication
5. Course structure
6. Grading

Team

**girafe
ai**

01

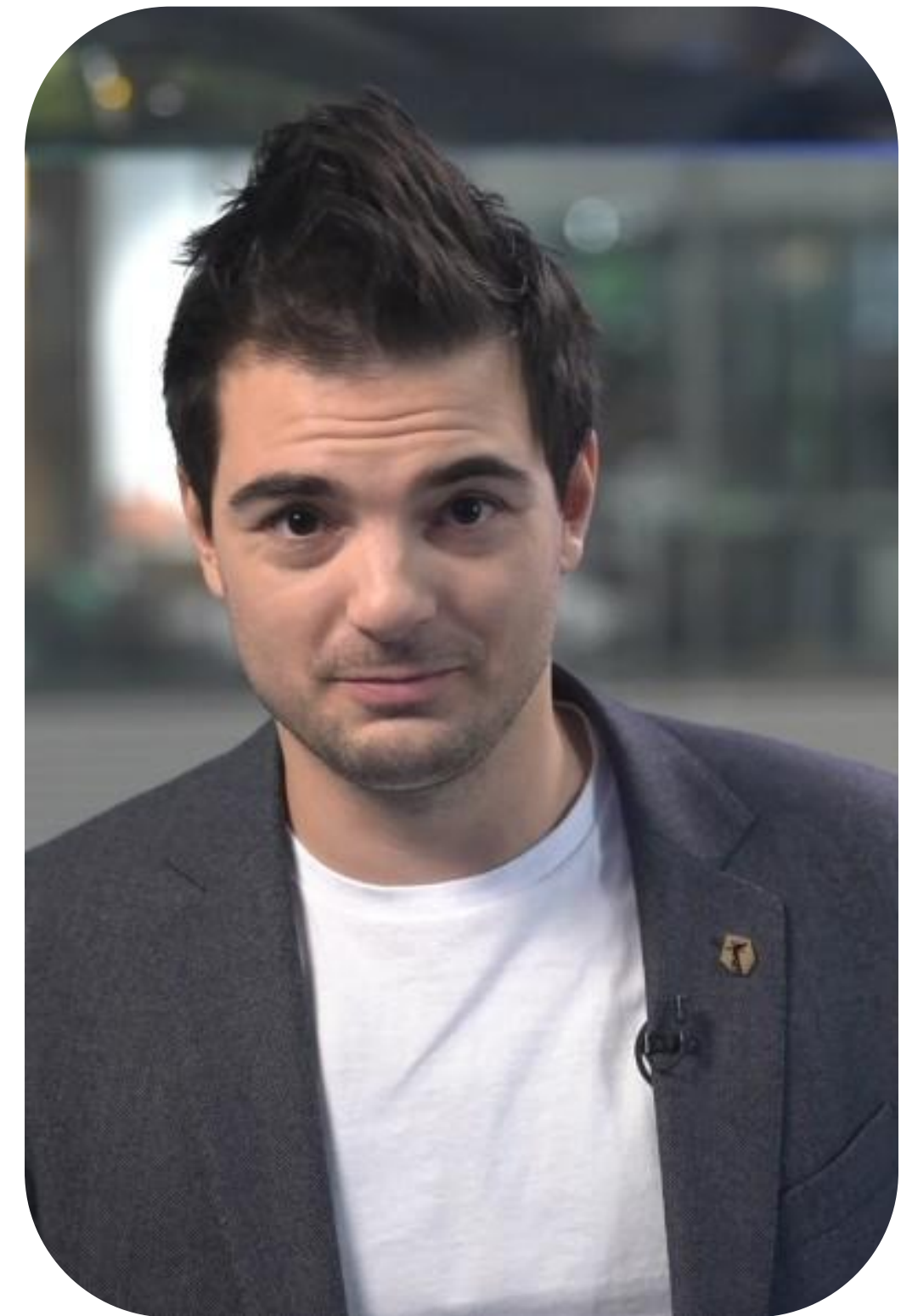
Vladislav Goncharenko

- Founder of girafe-ai
- Author of machine learning courses and Masters program at MIPT
- ML researcher (MIPT)
- Team lead of video ranking team at Dzen (yandex.ru)
- Ex-team lead of perception team at self-driving trucks (Evocargo)
- <https://www.linkedin.com/in/vladislav-goncharenko/>



Radoslav Neychev

- Founder of girafe-ai
- Author of machine learning courses at MIPT, MSU, and YSDA
- author of the Master's program "Modern Methods of Artificial Intelligence"
- Participated in the implementation of AI in experiments at the Large Hadron Collider.
- Areas of scientific interest: Artificial Intelligence; Large Language Models, Reinforcement Learning, Human Activity Recognition



Seminar instructors



Dmitry Kovrizhnykh

Teaches natural language processing — NLP, language models, and working with text data.



Arina Lapteva

ML Engineer and teacher at MIPT. Specializes in LLM/RAG technologies, agentic systems, and recommender systems.



Nikolay Karpachyov

Created courses in NLP and machine learning, teaches at MIPT, and works as a Deep Learning Engineer at Yandex. Researches generative text models.

Support team

- Assistants
 - grade homeworks
- Administrators
 - schedule, paperwork
 - @girafeai_support



Anastasiia
Galkina



Anastasiia
Orlova

How to learn?

girafe
ai

02

To learn or not to learn?

Why higher education matters

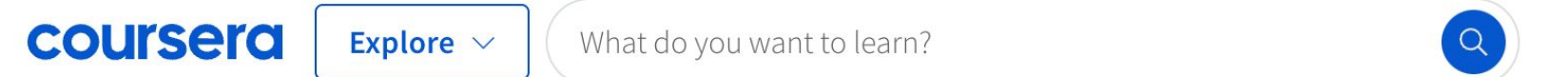
- Most published A* author in MSU
 - <https://habr.com/ru/articles/598305/>



— Чувак! А тебе не кажется, что сегодня в 21 веке намного эффективнее сразу пойти работать и, при желании, параллельно проходить курсы [ШАД](#), [Coursera](#), [Udacity](#), [Ozon-Masters](#) и т.д.? И высшее образование нафиг не нужно!

Science based learning techniques

- <https://www.youtube.com/watch?v=ddq8JIMhz7c>
- <https://www.youtube.com/watch?v=gBHpi2gyt94>
- <https://www.coursera.org/learn/learning-how-to-learn>
- https://wcer.wisc.edu/docs/resources/cesa2017/Dunlosky_SciAmMind.pdf
- https://www.uky.edu/~gmswan3/544/9_ways_to_reduce_CL.pdf



Home > Browse > Personal Development > Personal Development



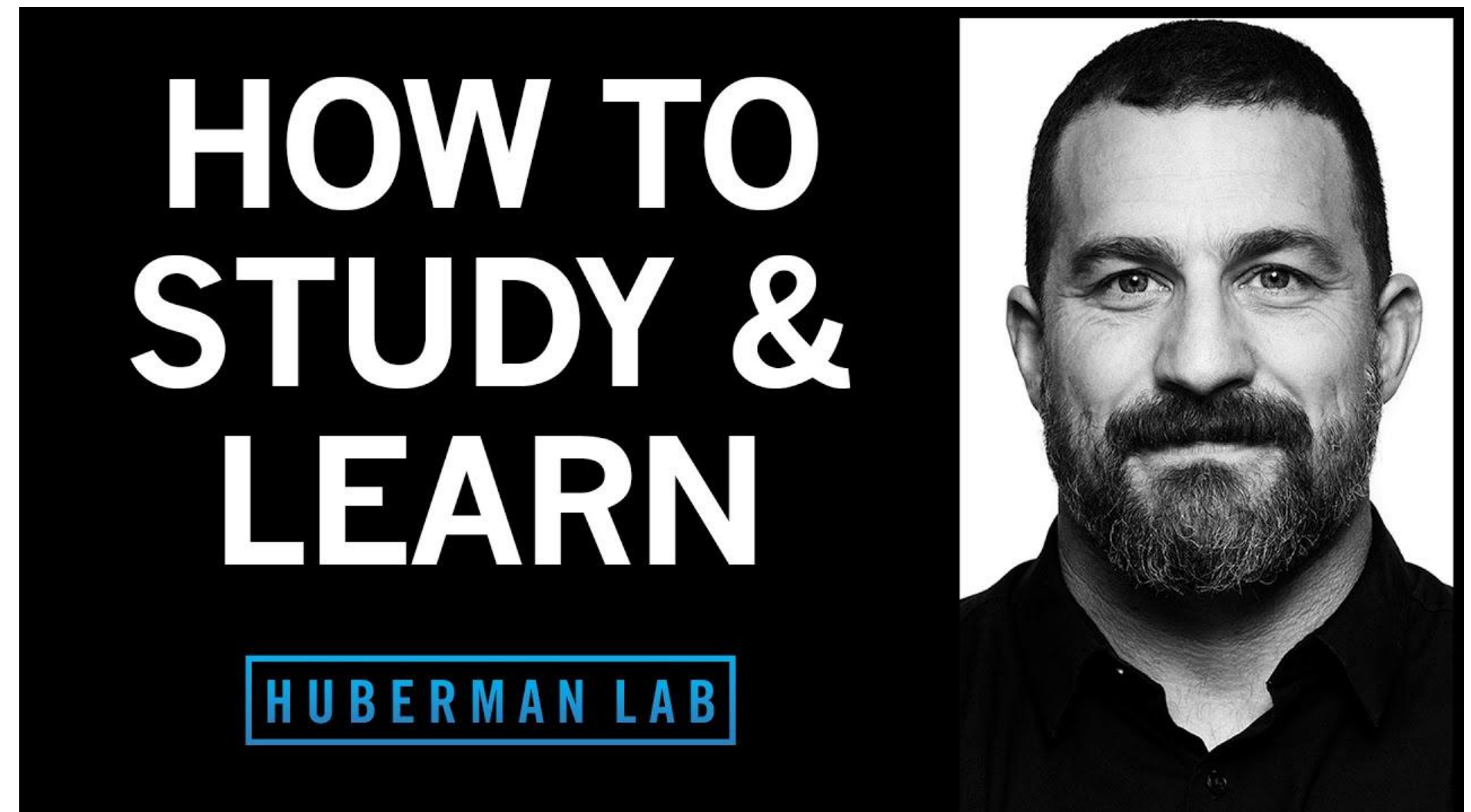
Learning How to Learn: Powerful mental tools to help you master tough subjects

🗣️ Taught in English | [22 languages available](#) | Some content may not be translated

Enroll for Free
Starts Sep 13

Financial aid available

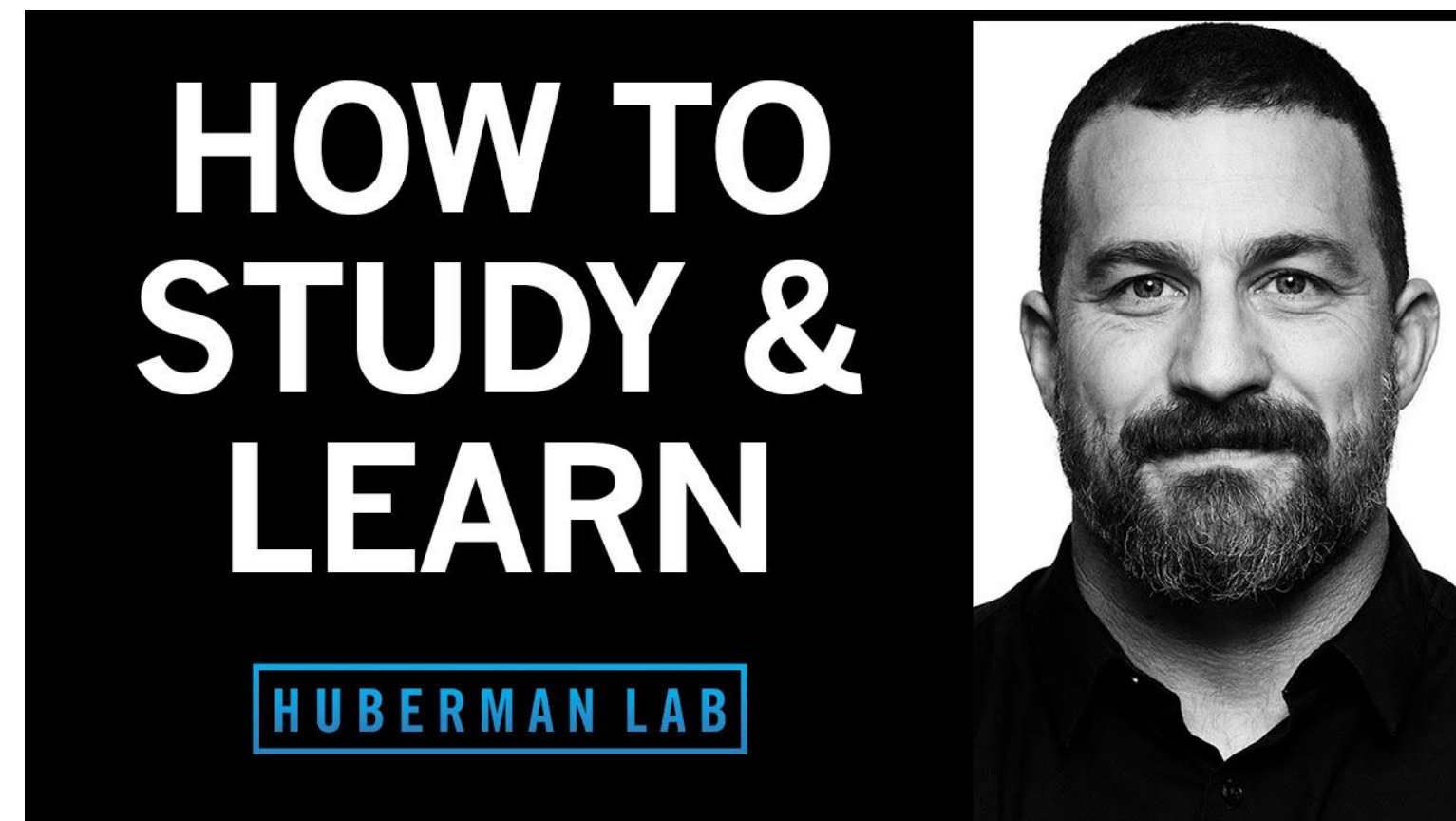
3,849,549 already enrolled



Science based learning techniques

Short outcomes:

- Reserve fixed time to study
- First study alone
- Then try to explain to your classmates
- Answer to open questions about the material by yourself



How to work with materials?

It is expected that **you need to study yourself** and no one will do it for you.

Machine Learning sphere is growing explosively and we try to give you the most during this course, so it is full of materials.

It is expected that you will need to **revisit each lecture and practical seminar** by your own. As well as look for some answers in different sources (internet, books, etc...)

Studying by other materials

It is not prohibited of course, but beware:

1. Search engine results first page
 - a. not structured at all (as a whole)
 - b. critically shallow
2. Chatbot answers
 - a. weak structure
 - b. superficial without advanced prompt
3. Other lectures, books, etc
 - a. structured differently
 - b. not identical in set of topics!

Exam will be held based on what actually thought in the course!

You have been warned!!!



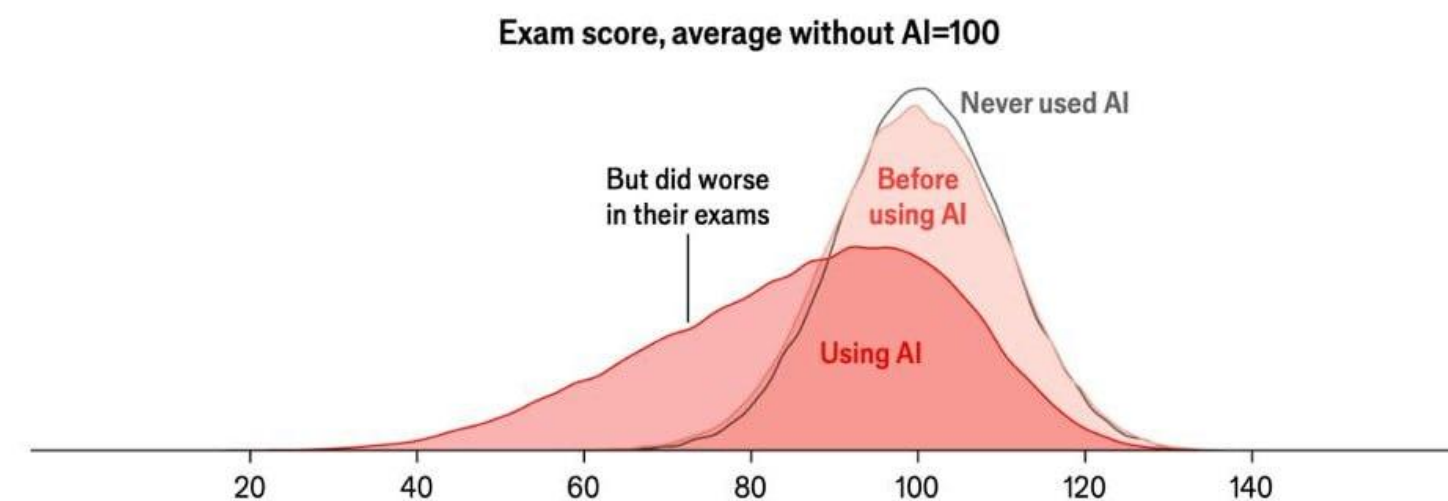
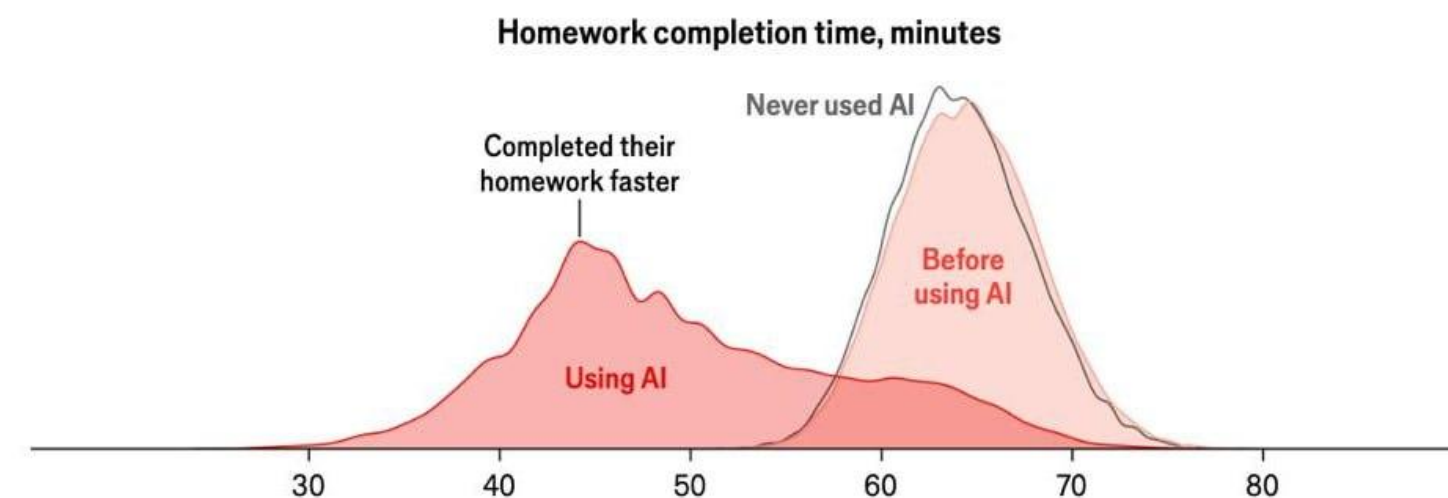
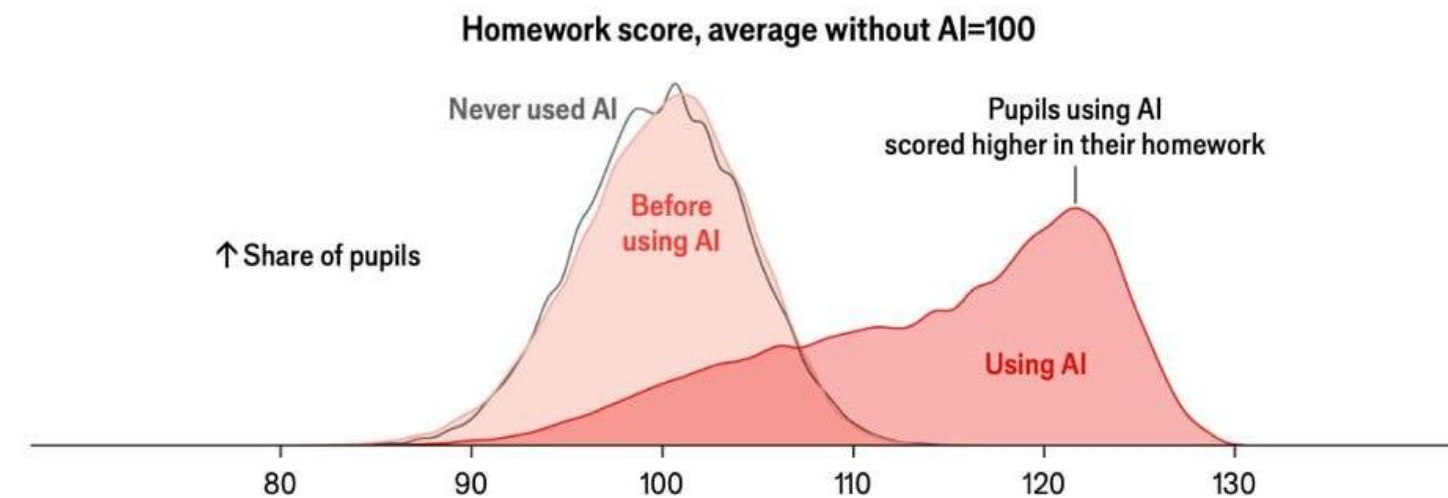
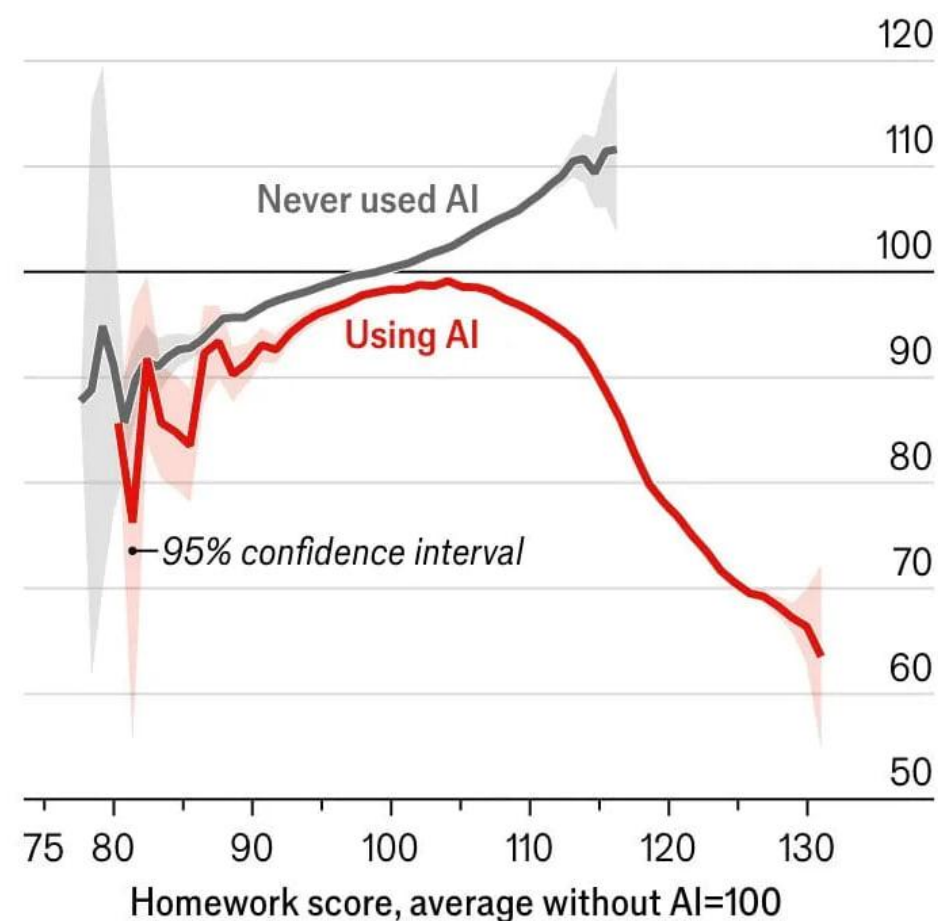
Learning with agents

- <https://www.skills.sh/mattpocock/skills/teach>
- <https://github.com/Egonex-AI/Understand-Anything>
- <https://t.me/vgcourse/564>

Thinking about it

China*, average exam score

Jan 2023-Jun 2025, average without AI=100



Source: "The generative AI learning penalty: evidence from Chinese secondary education", by D. Strömberg et al., preprint June 2026

*26,811 pupils aged 12-18

Prerequisites

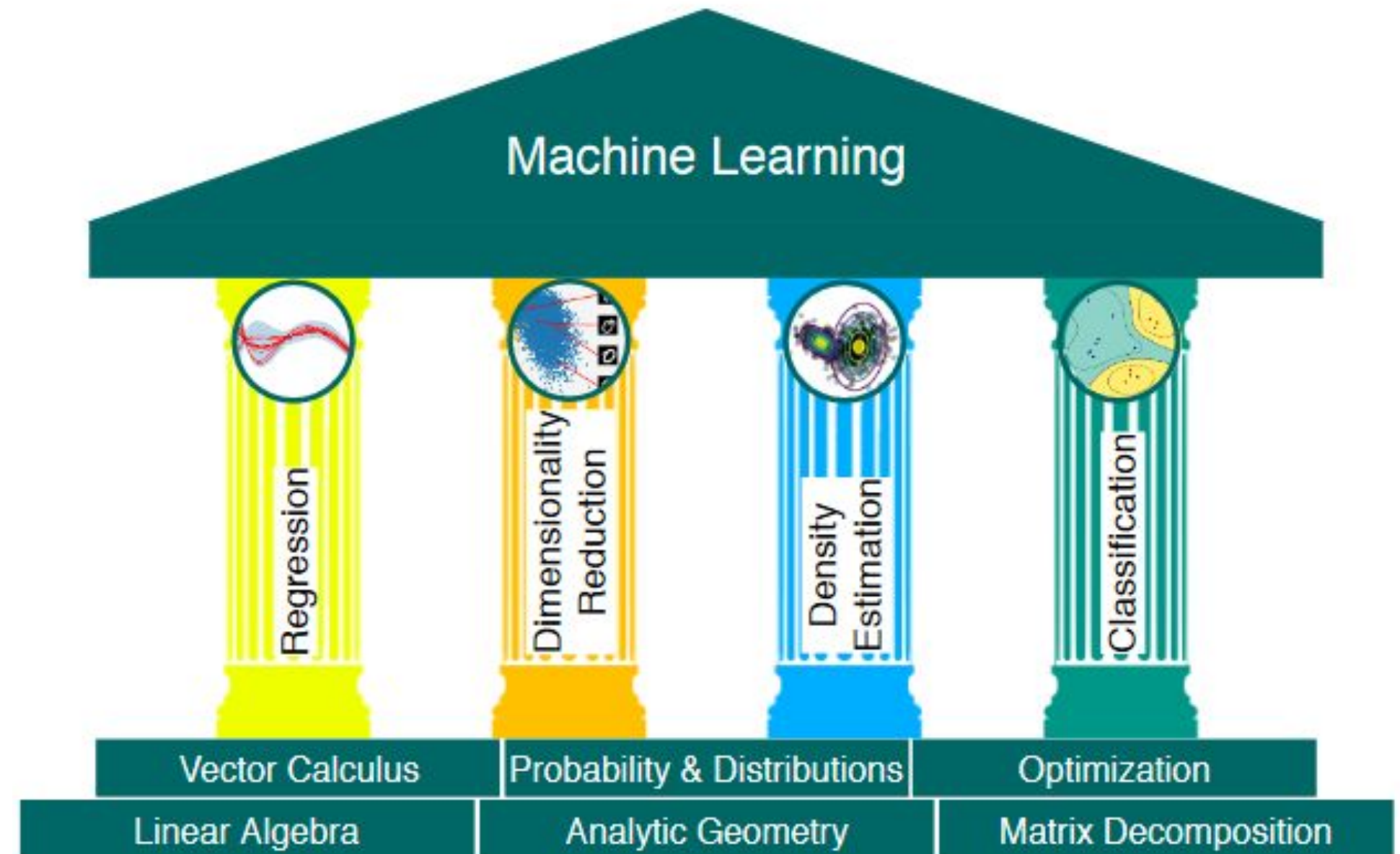
girafe
ai

03

Math requirements

Basic topics we expect you to know:

- Linear algebra
 - Matrices
 - Linear functions, transforms
 - Matrix decompositions
- Calculus
 - Matrix differentiation
- Statistics
 - Distributions
 - MLE
- Basics of optimization
 - Extremum
 - Convexity



Materials to refresh necessary math:

- <https://github.com/girafe-ai/math-basics-for-ai/>
- <https://mml-book.github.io/book/mml-book.pdf>

Math seminar

Revise linear algebra, calculus,
especially matrix differentiation.

Date will be announced in the chat.

$$\frac{\partial y}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial y}{\partial x_1} \\ \frac{\partial y}{\partial x_2} \\ \vdots \\ \frac{\partial y}{\partial x_n} \end{bmatrix}.$$

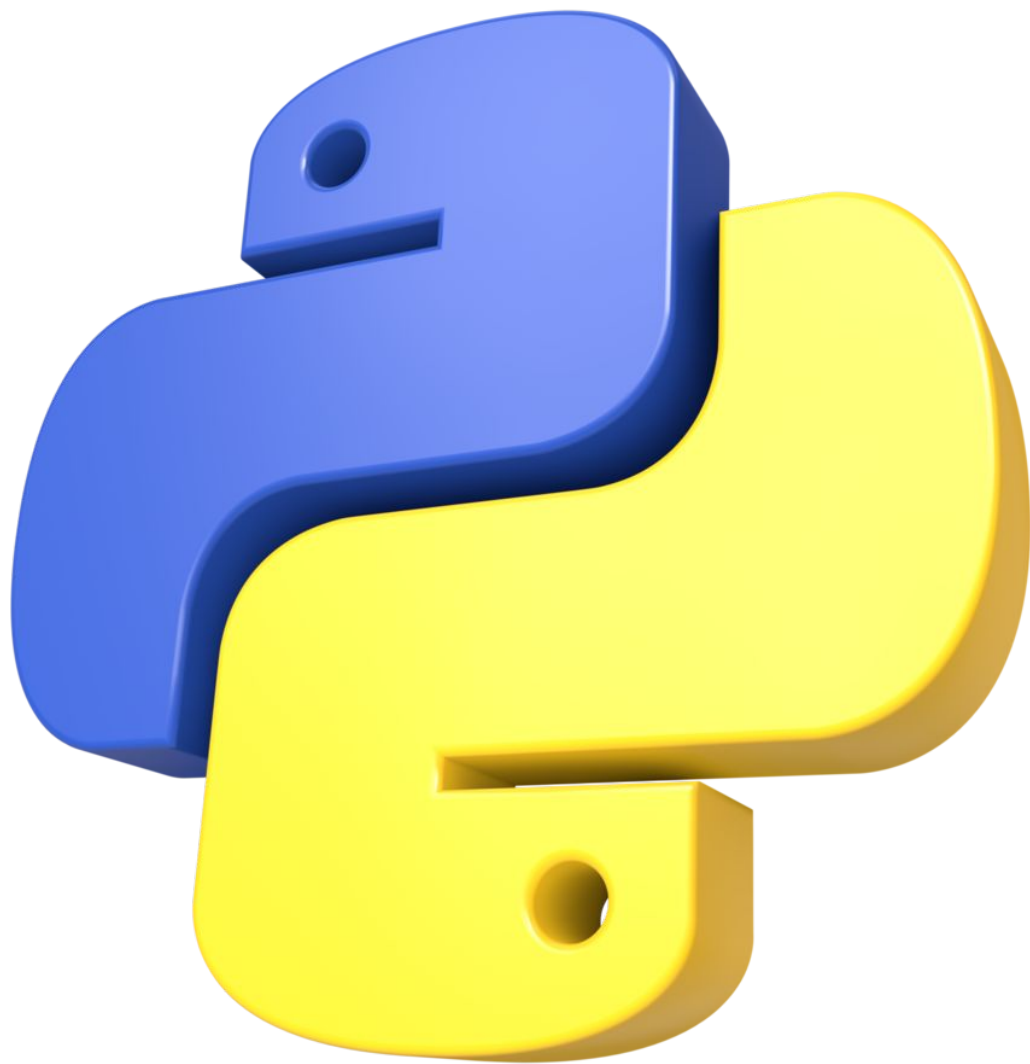
$$\frac{\partial \mathbf{y}}{\partial x} = \begin{bmatrix} \frac{\partial y_1}{\partial x} & \frac{\partial y_2}{\partial x} & \dots & \frac{\partial y_m}{\partial x} \end{bmatrix}.$$

$$\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_2}{\partial x_1} & \dots & \frac{\partial y_m}{\partial x_1} \\ \frac{\partial y_1}{\partial x_2} & \frac{\partial y_2}{\partial x_2} & \dots & \frac{\partial y_m}{\partial x_2} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y_1}{\partial x_n} & \frac{\partial y_2}{\partial x_n} & \dots & \frac{\partial y_m}{\partial x_n} \end{bmatrix}.$$

$$\frac{\partial y}{\partial \mathbf{X}} = \begin{bmatrix} \frac{\partial y}{\partial x_{11}} & \frac{\partial y}{\partial x_{12}} & \dots & \frac{\partial y}{\partial x_{1q}} \\ \frac{\partial y}{\partial x_{21}} & \frac{\partial y}{\partial x_{22}} & \dots & \frac{\partial y}{\partial x_{2q}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y}{\partial x_{p1}} & \frac{\partial y}{\partial x_{p2}} & \dots & \frac{\partial y}{\partial x_{pq}} \end{bmatrix}.$$



Programming requirements



Tools:

- Python, Numpy, Scipy
- Jupyter notebooks, VS Code
- UNIX terminal, git

Good places to study Python:

[4-hours long basic tutorial](https://snakify.org/en/)

<https://snakify.org/en/> or <https://pythontutor.ru/>

Computer Science basics:

<https://cs50.harvard.edu/x/>

And for UNIX suite:

<https://missing.csail.mit.edu/>

Some fundamentals

Conceptual and idiomatic dive into Linux command line:

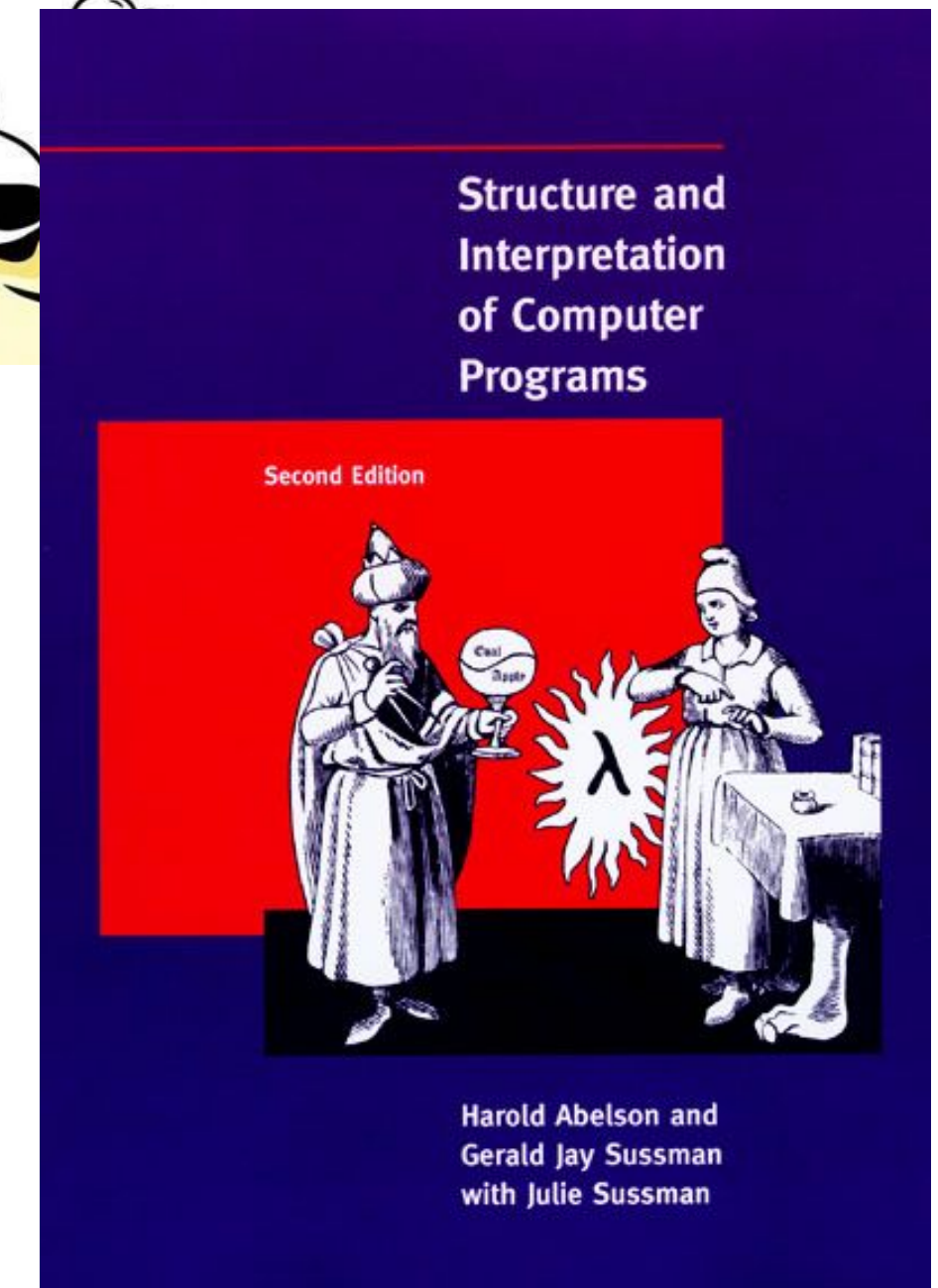
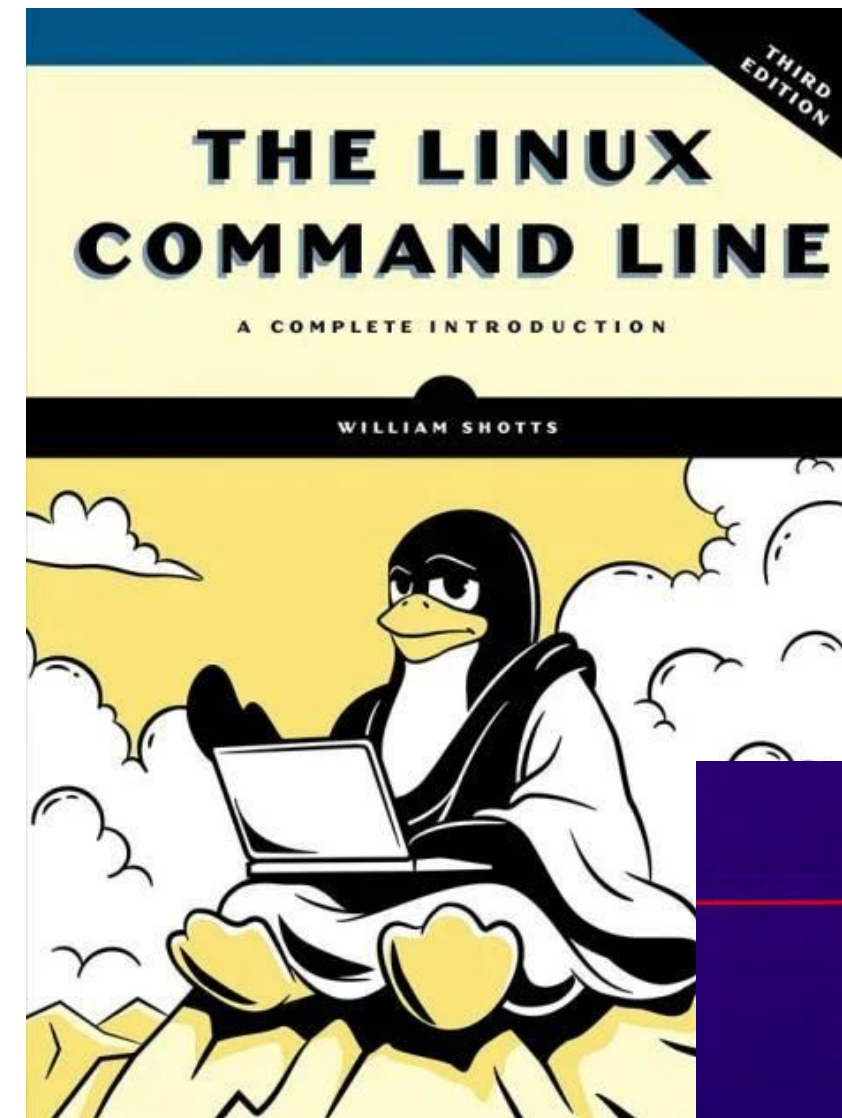
<https://www.linuxcommand.org/tlcl.php>

Fundamental introduction to programming:

[Structure and Interpretation of Computer Programs,](https://web.mit.edu/6.001/6.037/sicp.pdf)

<https://web.mit.edu/6.001/6.037/sicp.pdf>

[Code: The Hidden Language of Computer Hardware and Software](https://web.mit.edu/6.001/6.037/sicp.pdf)



Communication and materials

**girafe
ai**

04

Communication structure

Main communication is carried out by Telegram messenger.

1. Course channel
2. Course chat
3. Support account @girafeai_support telegram account

Probably **teacher will not answer to direct messages**
due to large number of students!



Lingua franca

Materials are provided in English because it is the language of international scientific communication thus all the new work is presented in English and we need to stay in line with it.

Also all top scientists in Russia publish all their significant works in English e.g. Ivan Oseledets

Materials

Are published mainly in course repository on github:

https://github.com/girafe-ai/ml-course/tree/itmo_26f

Also some materials can be send to Telegram chat

Videos are published in Youtube channel and Yandex disk.

We have both English and Russian recordings. If you need them, contact support account.



Course structure

girafe
ai

05

Basic course structure

1. Organisation. Introduction, MLE
2. Naive Bayes, Linear models
3. Validation cycle
4. Classification and LogReg
5. PCA, kNN, kNN indexes
6. Decision trees
7. Bias-Variance-Noise decomposition, inductive bias, Bagging, RF
8. Gradient Boosting done right. Contemporary libraries
9. Neural Networks: Linear layer, Activations, Backpropagation
10. RNN, LSTM. Inductive bias in NNs
11. Convolution operation. Classification architectures, datasets
12. MLM, Self-attention, Cross-attention mechanisms
13. Ranking
14. Representation learning
15. Unsupervised ML: intrinsic dim, manifold, etc, Geometric DL
16. LLM: decoder, encoder based architectures



Code of conduct

girafe
ai

06

Code of conduct

1. Do your own work
2. Acknowledge assistance
3. Use AI responsibly
4. Do not copy or share solutions
5. Report results honestly
6. Respect people and data
7. When in doubt, ask



Grading

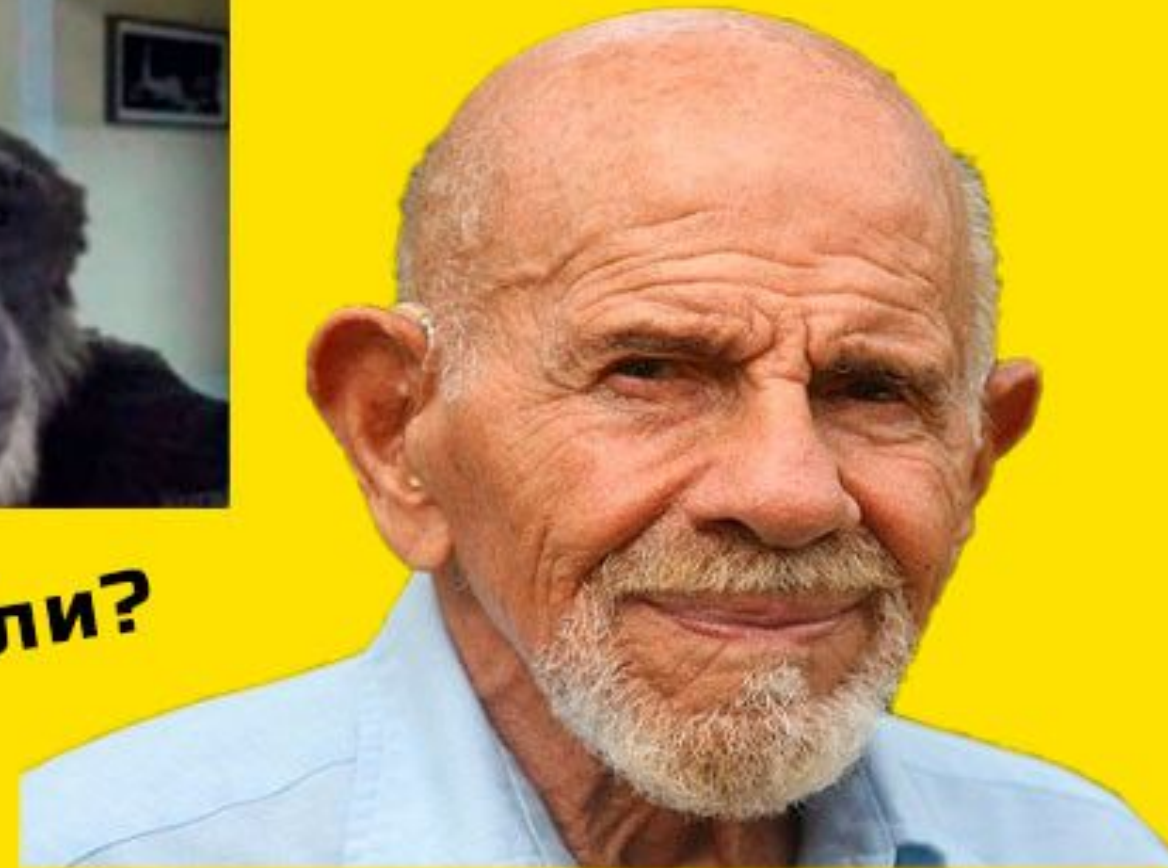
girafe
ai

07

Загадка от Жака Фреско



че задали?



на размышление даётся 30 секунд

Assignments and grading

Mark	Points	Activities		
		 Tests	 Contest	 Lab
уд	3	60	60	30
хор	5	75	70	50
отл	8	80	80	80

final_mark = min(tests, contests, labs)

Формат и автоматы

Формат

Оценка выставляется за работу в семестре, в конце прийти и закрыть всё за один день не получится.

“Автоматов” и перезачётов у нас нет.

В курсе есть ряд идей и подходов, которые нужно освоить и сдать посредством тестов, контестов, лабораторных



Дипломы с нами

Разные преподаватели имеют разные сферы научных интересов.

Если вы хотите делать проекты (в том числе дипломы) с кем-то - достаточно спросить этого человека.

Также если у вас уже есть тема то мы *возможно* сможем посоветовать кого-то в качестве потенциального научного руководителя.

Постараемся сделать курс для подготовки дипломов.

Список доступных тем опубликуем в канале курса



Thanks for attention!

Questions?

girafe
ai

